



maximum = factorial

Problem: Poisson Distribution Page No: 178 (E < X) 19
E = X

Arrival rate $\rightarrow \infty$.

$$P(X=0) = \frac{e^{-\lambda} \lambda^0}{0!} = 1$$

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Full factorial design, Basic concept of 2^2 , 2^3 and 2^k .

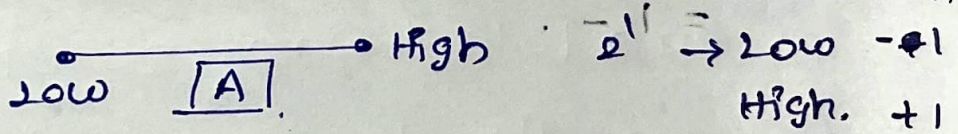
[(2,0) (2,0) 2,0,1 design]

* It is widely accepted that the most commonly used experimental designs in manufacturing

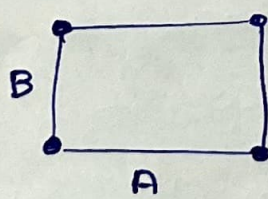
factor influencing in various levels - Factorial design

(X) Levels mentioned in sum (either low/high)

1 - factor

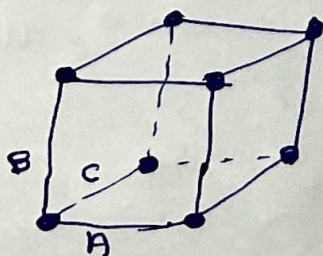


2 - factor



	A	B	AB
(i) A alone	-1	-1	+1
(ii) B alone	-1	+1	-1
(iii) Combined AB	+1	-1	-1
	+1	+1	+1

3 factors.



$$2^3 = 8$$

3 factors
 $2^3 \rightarrow 8$
level

Interaction

combination. A B C AB BC AC ABC

(1) $1 - 1 = 0$ to 1000 + 1000 = 2000

$\frac{1}{2} \times 98 = 49$

actant is involved? $\frac{B}{-} + \frac{1}{+} \rightarrow \frac{1}{-} + \frac{B}{+}$ $\frac{1}{-} + \frac{1}{+} \rightarrow \frac{1}{-} + \frac{1}{+}$ $\frac{1}{-} + \frac{1}{+} \rightarrow \frac{1}{-} + \frac{1}{+}$

BC -1 +1 +1 -1 +1 -1

A +1 -1 -1 -1 +1 . displacement

8. $\frac{1}{s^2(s^2+1)}$

AC $\frac{1}{s^2} \frac{1}{s^2+1} = \frac{1}{s^2} \left(\frac{1}{s^2+1} \right)$

AB

ABC

Balance and orthogonality

Each factors levels are tested equally (balanced) and the effects of each factor can be estimated independently of the others (orthogonal)

Balanced, in terms of factors

orthogonal : in terms of interactions.

Nolz

Interactions are also known as replications

Page No: 101. → Panneer selvam :

Eg: 4.8. The sales manager of Krishna Engg. company

wants to study the effect of sales regions as well as salesmen on quarterly sales revenue. So he has designed a factorial experiment involving 2 factors as shown in table. with 2 replications for each exp. combination.

		Salesman B			
		1	2	3	4
Sales region (A)	East	65	30	50	20
	West	70	85	65	40
	East	80	60	20	50
	West	85	55	25	40

As per this diagram each salesmen is assigned to each region for 2 different quarters. For ANOVA and check

the following at the significance level of 0.05.

(a) whether there is a significant difference in terms of sales revenue between sales regions

(b) whether there is a significant difference in terms of sales revenue b/w salesmen

(c) whether there is a significant diff in terms of revenue b/w diff. combinations of sales region and salesmen

Soln.

Salesmen

Region		Salesmen				sum, 2	$\sum x^2$	$\sum x^2/n$
		B ₁	B ₂	B ₃	B ₄			
A ₁		65	90	50	20	165		
		70	35	65	40	210		
A ₂		30	60	20	50	160		
		35	55	25	40	115		
x_{ij}								

$i \rightarrow$ Treatment of factor A (2)

$j \rightarrow$ Treatment of factor B (4)

$k \rightarrow$ Combinations.

Levels : 2

factor : 4.

Interactions : 2^4 .

$$Y_{ijk} = \mu + A_i + B_j + AB_{ij} + e_{ijk}.$$

μ - overall mean

The relationship b/w different sum of squares of this model is

$$\text{Total sum of squares} = (\text{sum of squares of the factor A}) + SS_B + SS_{AB} + SS_{\text{exp. err.}}$$

(SS_{tot})

$$SS_{\text{tot}} = SS_{\text{Reg.}} + SS_{\text{Salesmen}} + SS_{\text{Salesmen} \times \text{region}} + SS_{\text{error}}$$

SS_A :

$$SS_A = \sum_{i=1}^2 \left(\frac{Y_i^2}{8} - CF \right).$$

$k \rightarrow$ Combinations : $A_1B_1, A_1B_2, A_1B_3, A_1B_4$
 $A_2B_1, A_2B_2, A_2B_3, A_2B_4.$

Correction factor CF:

$$CF = \frac{(\sum x)^2}{n} = \frac{(\sum x_i)^2}{n}$$

$$SS_A = \frac{(\sum x_1)^2}{n_1} + \frac{(\sum x_2)^2}{n_2} + \frac{(\sum x_3)^2}{n_3} + \frac{(\sum x_4)^2}{n_4} - CF$$

$$SS_B = \frac{(\sum x_1)^2}{n_1} + \frac{(\sum x_2)^2}{n_2} - CF$$

$$SS_{AB} = \left(\sum_{i=1}^k \sum_{j=1}^t y_{ij}^2 \right) - CF$$

The relationship between sum of squares of two

ANOVA Table

source of variation

$$\left(\sum_{i=1}^k \sum_{j=1}^t y_{ij}^2 \right) - \frac{(\sum y_{ij})^2}{n} = A_{SS}$$

1. A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z